DT4012: Computer System Engineering I
SW Lab 1 Instructions:

1. The main objective of this lab is for you to practice using structs, arrays, basic control structures, and functions in C. All the information you need is covered in the first SW lecture.
2. You should work on this lab in groups of 2. You are not allowed to discuss or share your code with other class groups.

Good Luck.

1) Complex numbers are defined as numbers of the form $a + b\,i$, where a (the real part) and b (the imaginary part) are real numbers and $i^2 = -1$.

1. Define a structure type called complex with two fields to represent the real part and the imaginary part of a complex number. (Use a double to represent real numbers)
2. Write a function that takes as input a complex number z (use the structure you defined in a) and returns the conjugate zbar. (Hint: (1)The conjugate of $a + b\,i$ is $a - b\,i$, (2) return is the return value of the function not simply printing using printf).
3. Write four functions that take as input two complex numbers z1 and z2, and return z1+z2, z1-z2, z1*z2, and z1/z2. (Hint: revise how to do the multiplication and division of complex numbers before you attempt this question).
4. Write a function that takes as input a complex number z and returns the reciprocal of z.
5. Write a function that takes as input a complex number z and returns its magnitude (hint: the magnitude of a complex number $a + b\,i$ is $r = a^2 + b^2$, check math.h to compute the square root). Note that the magnitude of a complex number is a real number.
6. Write a main function that reads from the user two complex numbers (z1 and z2) and outputs z1+z1, z1-z2, z1*z2, z1/z2, and the magnitude and reciprocal of z1. (You can ask the user to input four real numbers representing the real and imaginary parts of z1 followed by the real and imaginary parts of z2 as a set of four real numbers).

2) In Appendix 1 two arrays of 100 numbers are listed which you will put in two vectors (copy and paste). The first array represents the real parts and the second array represents the imaginary parts of 100 complex numbers. The nth complex number is composed of the nth number from the first array as the real part and the nth number from the second array as the imaginary part.

Write functions that calculate the following properties of the vector.

a) Average magnitude (double) of those complex numbers
b) Minimum magnitude (double)
c) Maximum magnitude (double)
d) Variance in magnitude (double)
e) Write a main function that displays the values you computed in subparts a to d.

The input of all those functions should be an array of type complex and the size of the array (int). The output should be a double as it corresponds to the magnitude.

Appendix 1:

Array 1 (Real parts)

10.00000 5.000000 3.333333 2.500000 2.000000 1.666667 1.428571 1.250000 1.111111 1.000000 0.909091 0.833333 0.769231
0.714286 0.666667 0.625000 0.588235 0.555556 0.526316 0.500000 0.476190 0.454545 0.434783 0.416667 0.400000 0.384615
0.370370 0.357143 0.344828 0.333333 0.322581 0.312500 0.303030 0.294118 0.285714 0.277778 0.270270 0.263158 0.256410
0.250000 0.243902 0.238095 0.232558 0.227273 0.222222 0.217391 0.212766 0.208333 0.204082 0.200000 0.196078 0.192308
0.188679 0.185185 0.181818 0.178571 0.175439 0.172414 0.169492 0.166667 0.163934 0.161290 0.158730 0.156250 0.153846
0.151515 0.149254 0.147059 0.144928 0.142857 0.140845 0.138889 0.136986 0.135135 0.133333 0.131579 0.129870 0.128205
0.126582 0.125000 0.123457 0.121951 0.120482 0.119048 0.117647 0.116279 0.114943 0.113636 0.112360 0.111111 0.109890
0.108696 0.107527 0.106383 0.105263 0.104167 0.103093 0.102041 0.101010 0.100000

Array 2 (Imaginary parts)

7.000000 3.500000 2.333333 1.750000 1.400000 1.166667 1.000000 0.875000 0.777778 0.700000 0.636364 0.583333 0.538462
0.500000 0.466667 0.437500 0.411765 0.388889 0.368421 0.350000 0.333333 0.318182 0.304348 0.291667 0.280000 0.269231
0.259259 0.250000 0.241379 0.233333 0.225806 0.218750 0.212121 0.205882 0.200000 0.194444 0.189189 0.184211 0.179487
0.175000 0.170732 0.166667 0.162791 0.159091 0.155556 0.152174 0.148936 0.145833 0.142857 0.140000 0.137255 0.134615
0.132075 0.129630 0.127273 0.125000 0.122807 0.120690 0.118644 0.116667 0.114754 0.112903 0.111111 0.109375 0.107692
0.106061 0.104478 0.102941 0.101449 0.100000 0.098592 0.097222 0.095890 0.094595 0.093333 0.092105 0.090909 0.089744
0.088608 0.087500 0.086420 0.085366 0.084337 0.083333 0.082353 0.081395 0.080460 0.079545 0.078652 0.077778 0.076923
0.076087 0.075269 0.074468 0.073684 0.072917 0.072165 0.071429 0.070707 0.070000